

**ER DIAGRAM DESIGN AND
RELATIONAL SCHEMA**
Library Management System
Assignment 2

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OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Library Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ENTITIES AND ATTRIBUTES

Entity 1: MEMBER

Column	Key
Member ID	Primary Key
Member Name	-
Phone	-
Email	-
Address	-

Entity 2: LIBRARY_BRANCH

Column	Key
Branch ID	Primary Key
Branch Name	-
Phone	-
Address	-

Entity 3: BOOK

Column	Key
Book ID	Primary Key
Branch ID	Foreign Key
Title	-
Author	-
Genre	-
Availability	-

Entity 4: LOAN

Column	Key
Loan ID	Primary Key
Member ID	Foreign Key
Branch ID	Foreign Key
Loan Date	-
Due Date	-
Loan Status	-

Entity 5: LOAN_ITEM

Column	Key
Loan_Item_ID	Primary Key
Loan ID	Foreign Key
Book ID	Foreign Key
Quantity	-

Column	Key
Return_Date	-

Entity 6: FINE_PAYMENT

Column	Key
Payment_ID	Primary Key
Loan_ID	Foreign Key
Payment_Date	-
Payment_Method	-
Amount	-
Payment_Status	-

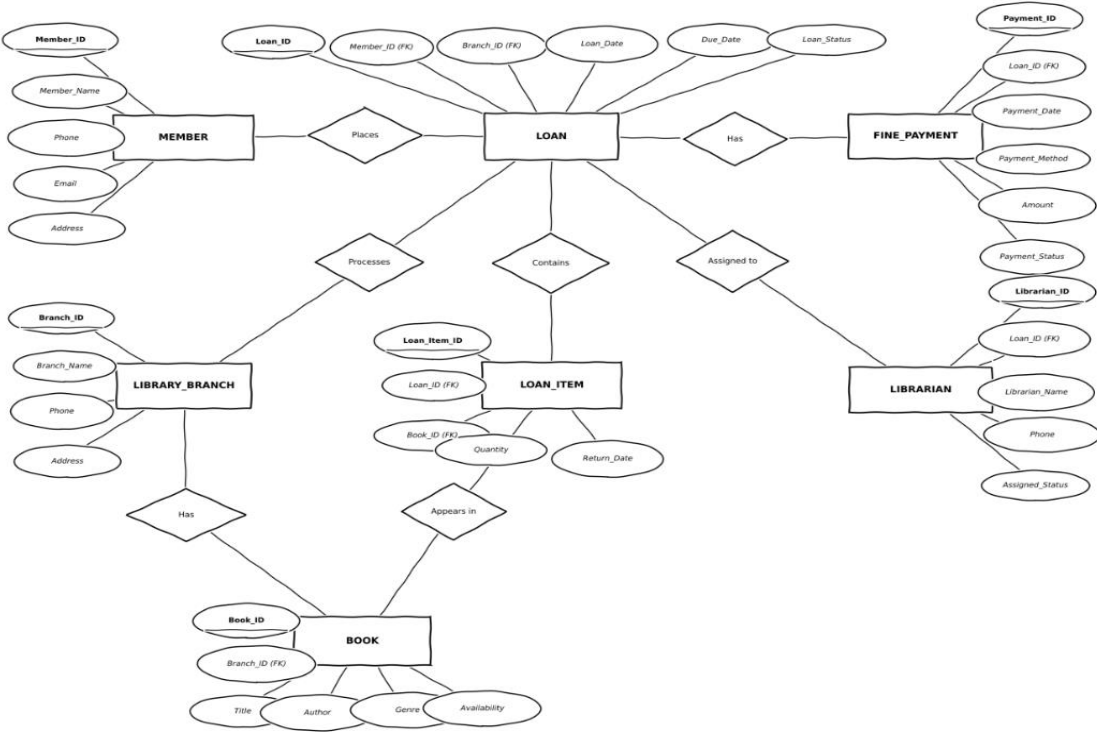
Entity 7: LIBRARIAN

Column	Key
Librarian_ID	Primary Key
Loan_ID	Foreign Key
Librarian_Name	-
Phone	-
Assigned_Status	-

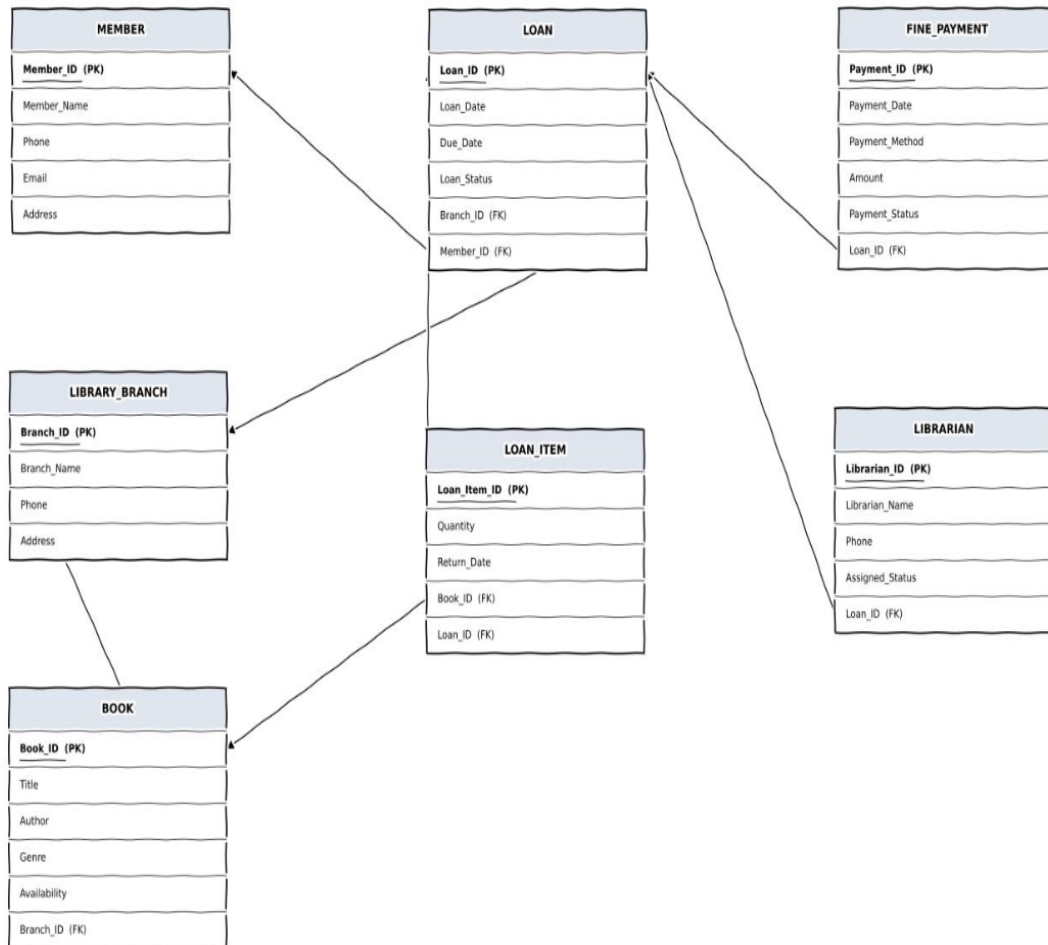
Relationships and Cardinalities

Entity 1	Relationship	Entity 2	Cardinality
MEMBER	Places	LOAN	1:M
LIBRARY_BRANCH	Has	BOOK	1:M
LIBRARY_BRANCH	Processes	LOAN	1:M
LOAN	Contains	LOAN_ITEM	1:M
BOOK	Appears in	LOAN_ITEM	1:M
LOAN	Has	FINE_PAYMENT	1:1
LOAN	Assigned to	LIBRARIAN	1:1

ER DIAGRAM



RELATIONAL SCHEMA



RELATIONAL SCHEMA — PRIMARY / FOREIGN KEYS

Table	Primary Key	Foreign Key
MEMBER	Member_ID	-
LIBRARY_BRANCH	Branch_ID	-
BOOK	Book_ID	Branch_ID
LOAN	Loan_ID	Member_ID, Branch_ID
LOAN_ITEM	Loan_Item_ID	Loan_ID, Book_ID
FINE_PAYMENT	Payment_ID	Loan_ID
LIBRARIAN	Librarian_ID	Loan_ID

RELATIONSHIP EXPLANATION

Member – Loan:

One member can place many loans. Therefore, the relationship is 1:M.

Library Branch – Book:

One library branch can hold many books. Therefore, the relationship is 1:M.

Library Branch – Loan:

One library branch can process many loans. Therefore, the relationship is 1:M.

Loan – Loan Item:

One loan can contain many loan items (books). Therefore, the relationship is 1:M.

Book – Loan Item:

One book can appear in many loan items over time. Therefore, the relationship is 1:M.

Loan – Fine Payment:

Each loan has at most one fine payment record. Therefore, the relationship is 1:1.

Loan – Librarian:

Each loan is assigned to one librarian record. Therefore, the relationship is 1:1.

